DeepScientist

AI REVIEW REPORT

AIG_CUR_2026_FARS_token_balanced

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Document No.	6390b3e1-5300-4cde-9bf7-7287edfbae08	Work ID	DS-RV-6390B3E15
Status	Completed	Agent Model	gpt-5.2
Token Usage	Input 12049919 | Output 224864 | Total 12274783	LLM Requests	74
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Final Agent Report

Structured markdown synthesized by the annotation agent.

Summary

This paper studies the "Brain Rot" phenomenon reported for continual pretraining (CPT) on short, low-quality social media text, and argues that a key driver is an *implicit token-weight imbalance* induced by per-sequence loss reduction: short sequences receive larger effective per-token gradient weight (Page 1 - Abstract; Page 2–3 - Method: Eq. (1)–(4)). The proposed mitigation is "token-balanced packing," i.e., concatenating short sequences into fixed-length blocks to equalize per-token contribution while also improving throughput (Page 3 - Method: Token-Balanced Packing; Page 5 - Experiments: Training Efficiency).

Empirically, under a 3-condition design (control/packed vs junk/packed vs junk/unpacked), the paper reports that packing largely closes the gap between packed and unpacked junk CPT on ARC-CoT and RULER, and yields large wall-clock gains (Page 4–5 - Experiments: Main Results; Page 5 - Training Efficiency). The direction of the main result (packed junk ≈ packed control ≫ unpacked junk) is consistent with a training-dynamics explanation and is practically interesting.

However, the current manuscript over-claims causality and "safety." Several validity-critical details are missing or internally inconsistent: (i) the mechanism depends on the *actual loss reduction implementation* (per-sequence vs per-token), which is not specified and is not universal across LLM training stacks; (ii) the headline 8.17× disparity is not transparently derived from the stated mean lengths; (iii) packing changes optimizer-step counts and sequence composition, so the A/B/C design does not yet isolate token-weight disparity as the unique cause; (iv) evaluation protocol (prompts/decoding/scoring) and training hyperparameters are underspecified; and (v) the RULER Variable Tracking sub-task reverses the trend (unpacked best), which needs diagnosis before broad "long-context recovery" claims (Page 6 - Experiments: Variable Tracking anomaly).

Overall, the paper has a promising, high-leverage practical hypothesis, but it needs tighter math-to-implementation alignment, stronger causal controls/ablations, and bounded, evidence-consistent wording before the central "not semantic / safe" conclusions are defensible.

Strengths

- **Clear, practically relevant hypothesis with a simple mitigation**: The paper articulates a concrete training-dynamics hypothesis (length-induced implicit token-weight imbalance under per-sequence loss) and proposes an easy-to-adopt mitigation (packing to fixed length) that practitioners already care about for throughput (Page 1 - Introduction; Page 3 - Method: Token-Balanced Packing).
- **Controlled experimental design that directly targets the semantic-vs-dynamics debate**: The A/B/C conditions are a good starting point for separating "junk content" from "packing" (Page 3 - Experimental Design; Page 4 - Main Results).

- **Large effect size on the key comparison (packed vs unpacked junk)**: The reported gap between Condition B and C is large on both ARC-CoT and RULER, and the paper reports multi-seed averages and significance tests (Page 4–5 - Main Results).
- **Strong efficiency motivation and substantial measured throughput gains**: The speedup claims, if normalized appropriately, would be a strong practical contribution: fewer steps, higher tokens/sec, and much shorter wall-clock CPT time (Page 5 - Training Efficiency).
- **Good transparency signals**: Explicit code availability and an explicit automated-generation disclaimer are unusual but helpful; with a stronger reproducibility checklist, these can increase reviewer trust (Page 1 - Disclaimer).
- **Failure-mode surfacing (partially)**: The RULER sub-task breakdown exposes that recovery is not uniform, and surfaces at least one contradictory sub-task (Variable Tracking) that can guide future analysis (Page 5–6 - RULER Sub-task Analysis).

Weaknesses

- **Validity-critical mechanism depends on an unspecified implementation detail (loss reduction)**: The core argument assumes per-sequence averaging (Eq. (1) then Eq. (2)) so that per-token weights scale like $1/L_i$ (Page 2–3 - Method). Many LLM stacks compute per-token batch means, which would largely remove the claimed imbalance even without packing. The paper must state and verify the actual reduction used in experiments, otherwise the causal mechanism is not established.
- **Numeric and definitional inconsistencies weaken trust**:
  - The headline 8.17× disparity is not transparently derived from the stated mean lengths (16.7 vs 100.4 implies ~6×), suggesting the wrong "mean" is being used or the computation is unclear (Page 3 - Method: 8.17× example).
  - "Recovery relative to no-CPT baseline" appears in Abstract/Conclusion, but the recovery definition is actually relative to the A–C gap; ARC recovery >100% is therefore easy to misinterpret (Page 1 - Abstract; Page 4 - Main Results).
- **Causal identification is under-controlled because packing changes multiple factors**: Packing changes (i) token-weighting under per-sequence loss, (ii) sequence composition / cross-document context exposure, and (iii) the number of optimizer updates for a fixed token budget. The A/B/C design does not yet isolate per-token weighting as the unique cause without additional controls (Page 3 - Experimental Design; Page 5 - Mechanism Analysis).
- **Reproducibility and fairness details are insufficient**: Key training details are missing (token budget, steps, LR schedule, batch tokens, grad accumulation, masking/position-id scheme for packing). Without these, readers cannot reproduce results or judge fairness (Page 4 - Setup; Page 3 - Packing details).
- **Long-context evidence has an unresolved contradiction**: In the RULER sub-task breakdown, Variable Tracking reverses the trend (unpacked best; packed worst) and is dismissed as "anomalous" without diagnosis. This directly limits broad claims about "safe" long-context recovery (Page 6 - RULER anomaly).
- **Over-strong claims and scope creep**: Statements like "semantic content does not inherently harm reasoning" and "CPT on short-sequence data is safe" exceed the evidence (single model size, single domain, incomplete controls; packed conditions below no-CPT baseline on RULER) and should be bounded (Page 5 - Main Results interpretation; Page 6 - Conclusion).
- **Related work positioning is under-developed**: The related-work section reads as a broad list and does not clearly position against nearest-neighbor work on packing

correctness/cross-contamination and on sequence composition effects from concatenation/packing, nor against token weighting approaches; it also mixes self-claims into related work (Page 2 - Related Work).

Key Issues

Ranked Error Board (Top-5)

1. Mechanism–implementation mismatch (loss reduction not specified/verified)
   - **Anchor**: Page 2–3 - Method: Eq. (1)–(3) assumes per-sequence reduction.
   - **Risk**: If actual training uses per-token averaging, the claimed token-weight disparity (and thus causal explanation) may not hold.
   - **Fixability**: High (report code + add one control run).
   - **Confidence**: High.
2. **Causal confounding: packing changes steps + sequence composition, not only token weights**
   - **Anchor**: Page 3 - Experimental Design; Page 5 - Training Efficiency implies huge step-count changes.
   - **Risk**: The observed recovery could be due to matched token budgets vs step budgets, or due to cross-document context effects, undermining the "root cause" claim.
   - **Fixability**: Medium (add 2–3 targeted ablations).
   - **Confidence**: High.
3. Quantitative inconsistencies and ambiguous recovery framing (8.17×; baseline-relative recovery)
   - **Anchor**: Page 3 - "mean length 16.7 vs 100.4 → 8.17×" inconsistency; Page 1/4 - recovery definition and wording mismatch.
   - **Risk**: Undermines trust and can be interpreted as sloppy calculation even if directionally correct.
   - **Fixability**: High (clarify formula; report harmonic means / E[1/L]; rewrite abstract/conclusion).
   - **Confidence**: High.
4. **Long-context claim is weakened by an unresolved contradictory sub-task**
   - **Anchor**: Page 6 - Variable Tracking anomaly (C≫A,B).
   - **Risk**: Broad "safe" / "long-context recovery" claims may be rejected; could indicate evaluation bug or masking side effect.
   - **Fixability**: Medium (audit scripts; add diagnostics; bound claims).
   - **Confidence**: Medium–High.
5. **Reproducibility & fairness reporting is below the bar for a causal-training-dynamics claim**
   - **Anchor**: Page 4 - Setup lacks token/step budgets, LR schedule, batch tokens, packing mask details, evaluation prompts.
   - **Risk**: Readers cannot verify that results are not due to hidden knobs; prevents adoption.
   - **Fixability**: High (add config tables + appendix).
   - **Confidence**: High.

Page Coverage Audit

Page	Annotation Count	Coverage Status	Skip Reason
1	6	covered	—

Page	Annotation Count	Coverage Status	Skip Reason
2	2	covered	—
3	4	covered	—
4	2	covered	—
5	4	covered	—
6	3	covered	—
7	0	skipped	References-only page

Actionable Suggestions

Must (publication-critical)

1. **Specify and verify loss reduction + masking exactly (and align math to code)**
   - **Where**: Page 2–3 - Method (Eq. 1–4); Page 4 - Setup.
   - **Do**: Add a subsection "Loss reduction and masking implementation" with pseudocode:
     - how you compute token-level loss with attention masks,
     - whether you average per sequence or per token,
     - treatment of padding and EOS,
     - for packing: block-diagonal attention vs naive causal attention, and whether position ids reset per original sample.
   - **Acceptance criteria**: A reader can reproduce the exact weight computation that yields 8.17× and can implement your packing without cross-contamination.
2. Add 2–3 targeted ablations to isolate the causal factor
   - **Ablation A (weight only)**: Unpacked junk CPT with **per-token batch-mean loss** (removes length-induced weighting without packing).
   - **Ablation B (composition only)**: Packed CPT with **strict block-diagonal mask** vs **naive cross-document causal mask**.
   - **Ablation C (step-count control)**: Match optimizer steps (via grad accumulation / batch tokens) between packed and unpacked at matched token budgets.
   - **Acceptance criteria**: You can state which factor(s) account for most of the degradation, and you can downgrade claims if results are mixed.
3. Fix quantitative framing and baseline language (avoid misleading "safe" and "recovery vs no-CPT")
   - **Where**: Page 1 - Abstract; Page 4 - Main Results; Page 6 - Conclusion.
   - **Do**: Use one consistent definition of "recovery" (explicitly A–C gap) and separately report deltas vs no-CPT. Replace universal claims with bounded statements.
4. **Diagnose the Variable Tracking reversal rather than calling it anomalous**
   - **Where**: Page 6 - RULER analysis.
   - **Do**: Audit evaluation parsing, provide qualitative examples, test prompt/context length sensitivity, and if it persists, present it as a failure mode tied to masking/composition.
5. **Upgrade reproducibility reporting to match the strength of the claims**
   - **Where**: Page 4 - Setup + Appendix (recommended).
   - **Do**: Add a compact config table: token budget, steps, epochs, LR schedule, optimizer, batch tokens, grad accumulation, weight decay, clipping, packing parameters; publish

exact prompts/decoding for ARC and RULER.

Nice-to-have (high ROI but optional)

6. **Cross-domain / cross-model validation**: replicate on (i) another model size or family, and/or (ii) another short-text domain with similar length distribution.
7. **Mechanism-level analysis aligned with Brain Rot forensics**: measure "thought-skipping" style errors (e.g., truncated CoT) as in [1], and test whether packing reduces them.
8. **Related-work restructuring**: reorganize by axes (packing correctness [2][4], sequence composition [3], explicit token weighting [6]) and state explicit differences.

One copy-ready rewrite (Abstract safety claim)
Mentor Revised Version (replace last abstract sentence):
"Our results suggest that, under the tested `CPT→instruction-tuning` pipeline and with correct packing/masking, short-sequence CPT does not exhibit the severe degradation seen in the unpacked setting; broader safety claims require validation across models, corpora, and loss-reduction implementations."

Storyline Options + Writing Outlines

Option A — "Mechanism-first, then controlled falsification" (recommended)

1. Start with the operational definition of Brain Rot (which benchmarks drop, under what `CPT→IT` pipeline).
2. State the competing hypotheses (semantic-quality vs training-dynamics) and the falsifiable prediction pattern ($A \approx B \gg C$ vs $A \gg B \approx C$).
3. Introduce the mechanism: per-sequence reduction induces length-dependent token weighting; packing removes it (with correct masking).
4. Preview the key empirical pattern + the main caveats/controls (loss reduction verification, step-count control, masking).
5. End with 3 crisp contributions.

Option B — "Practitioner pain point: safe short-text CPT"

1. Lead with the practical need: adapting to fast-moving short-text domains without losing reasoning/long-context.
2. Describe the observed failure mode (Brain Rot) and why current advice (filtering/avoid short text) is costly.
3. Propose a simple pipeline fix (token-balanced packing + masking) and preview that it mitigates degradation + speeds training.
4. Immediately bound claims: mitigation under tested protocol; broader safety pending.

Option C — "Packing is not just efficiency"

1. Lead with packing's ubiquity for throughput.
2. Point out an under-discussed side effect: loss reduction + variable length can create implicit token reweighting; sequence composition can also change context distribution.
3. Show that these training-dynamics details can materially change capability retention in CPT.

Abstract Outline (complete; 4–5 sentences)
S1 (Problem + setting): Define Brain Rot operationally as capability drops observed after CPT on short social-media text under a `CPT→instruction-tuning` pipeline (Page 1 -

Abstract; Page 4 - Setup).
S2 (Gap): Note that prior explanations emphasize semantic data quality, but training procedures for variable-length short sequences can introduce systematic optimization biases (Page 1 - Introduction; Page 2–3 - Method).
S3 (Method idea): State the mechanism (per-sequence loss reduction upweights short sequences at the token level) and the intervention (token-balanced packing with correct masking to equalize token contribution) (Page 3 - Eq. (2)–(4)).
S4 (Evidence): Summarize the controlled A/B/C result pattern (packed junk ≈ packed control ≫ unpacked junk) on ARC-Challenge and RULER, with the recovery definition clearly scoped to the A–C gap (Page 4 - Main Results).
S5 (Implication + bounds): Conclude that results are consistent with a training-dynamics contributor to Brain Rot and that packing is a practical mitigation; explicitly bound generality (single model/domain; loss reduction + masking controls needed for broad "safe" claims) (Page 6 - Conclusion).

Introduction Outline (complete; paragraph-by-paragraph)
P1 (Territory + operational definition): Define CPT and why practitioners use it; define Brain Rot with specific evaluation signals (ARC-CoT, RULER@4K) and the `CPT→IT` protocol used in the cited setup. Transition: "This raises the question of what causes the degradation."

P2 (Competing explanations + gap): Summarize the prevailing semantic-quality explanation and its practical consequence (filter/avoid short text). State the gap: variable-length training introduces procedural biases not accounted for. Transition: "We test whether an optimization artifact can explain the effect."

P3 (Core mechanism, with assumptions stated): Introduce per-sequence vs per-token loss reduction and explicitly state that the hypothesized imbalance exists under per-sequence reduction; preview the computed disparity and how it is measured (E[1/L] / harmonic mean). Transition: "This suggests a simple mitigation."

P4 (Solution + correctness constraints): Present token-balanced packing, but immediately clarify the correctness requirements (masking/position ids; no cross-contamination unless intended). Acknowledge that packing can change step counts and sequence composition, so controls are needed. Transition: "We therefore use a controlled design."

P5 (Evidence preview + falsifiable prediction): Describe Conditions A/B/C and the predicted pattern under each hypothesis. State what is held constant (token budget/steps) and what controls are added (token-avg loss ablation; strict vs naive packing). Transition: "We evaluate reasoning and long-context."

P6 (Results preview + bounded takeaway): Give one sentence on the main observed pattern + one sentence on the main caveat/failure mode (Variable Tracking reversal). Transition: "We summarize contributions."

P7 (Contributions list): Provide exactly three contributions (mechanism, method/implementation, controlled evidence + efficiency), each with scope bounds.

(Recommended writing change) Throughout intro/abstract, replace strong causal words ("demonstrate … not semantic", "safe") with evidence-consistent language unless the added ablations are completed.

Priority Revision Plan

P0 — Must fix for validity/defensibility (1–3 days)

1. **Lock down the training objective details** (Page 2–3 - Method; Page 4 - Setup)
   - Add explicit formulas for both per-sequence and per-token reductions and state which one you used.
   - Add a code/pseudocode snippet + a one-line unit test that reproduces the 8.17× weight ratio from dataset statistics.
2. **Fix all baseline/recovery language + numeric consistency** (Page 1 - Abstract; Page 4 - Main Results; Page 6 - Conclusion)
   - Use one consistent recovery definition (A–C gap) and avoid "relative to no-CPT baseline" unless you recompute.
   - Clarify why ARC recovery can exceed 100%.
   - Replace "safe" and "not semantic" with bounded statements.
3. **Add minimal causal controls (one table + 2 ablations)** (Page 3 - Experimental Design; Page 5 - Mechanism)
   - Add a "tokens/steps/batch-tokens/LR" table per condition.
   - Run: (i) Unpacked + per-token loss reduction; (ii) Packed strict mask vs packed naive mask (or explicitly demonstrate you already used strict mask).
4. **Resolve Variable Tracking anomaly** (Page 6)
   - Audit evaluation parsing and report qualitative failures.
   - If it persists, bound claims and list it as a known failure mode.

P1 — Strengthen novelty positioning + reproducibility (3–7 days)

5. **Related-work rewrite into a taxonomy + nearest-neighbor comparisons** (Page 2)
   - Add packing correctness/cross-contamination references and explicitly state what is new in your claim.
   - Add sequence-composition discussion and explain why it is not the primary mechanism (or measure it).
6. **Reproducibility appendix + artifact checklist**
   - Provide exact commit hash, dataset manifests/checksums, training commands, prompts, and scoring scripts.
7. **Evaluation robustness**
   - Add one longer-context point for RULER if feasible (e.g., 8K/16K) and one prompt-variant check for ARC-CoT.

P2 — Broader impact and generalization (1–3 weeks)

8. **Cross-model / cross-domain replication**
   - Replicate on a second model size/family and/or a second short-text domain with similar length distribution.
9. **Mechanism-level bridging to Brain Rot forensics**
   - Measure "thought-skipping" (e.g., truncated reasoning traces) and show packing reduces it, to connect to [1].

ASCII Diagram — Revision Strategy Roadmap

[Over-strong claim: 'safe' + 'not semantic']
-> [Evidence gap: objective impl + confounds not controlled]
-> [Risk: novelty/validity doubts]
-> [Fix P0: specify loss reduction + masking + ablations + rewrite claims]
-> [Expected gain: causal attribution defensible]

[Mechanism number: 8.17× unclear]
-> [Fix P0: derive via E[1/L], report arithmetic+harmonic means]
-> [Expected gain: trust + clarity]

[RULER Variable Tracking reversal]
-> [Fix P0: eval audit + failure-case analysis]
-> [Expected gain: bounded long-context claims]

[Related work list-style]
-> [Fix P1: taxonomy + head-to-head deltas vs packing/sequence-composition/ token-weighting work]
-> [Expected gain: novelty positioning]

Experiment Inventory & Research Experiment Plan

Exp ID	Objective/Hypothesis	Setup (data/split/protocol/baselines)	Metrics	Main Outcome	Claim Supported	Current Limitation
E1	Test whether packing mitigates Brain Rot degradation on junk short text	Base: Llama-3-8B-Instruct; CPT on junk tweets; compare **packed vs unpacked**; then instruction tune on Alpaca-5k; 3 seeds	ARC-Challenge (CoT), RULER@4K	Packed junk ≫ unpacked junk on both metrics	Packing mitigates the previously reported degradation in this protocol	Token/step budgets and loss reduction implementation not fully specified; packing may change composition + steps
E2	Test whether "semantic quality" is necessary for degradation	Compare **packed junk** vs **packed control** CPT, then same IT	ARC-CoT, RULER@4K	Packed junk ≈ packed control; unpacked junk much worse	Degradation not solely explained by junk content under this protocol	Still single model/domain; packing changes more than token weighting; packed conditions below no-CPT on RULER
E3	Quantify mechanism proxy (token-weight disparity)	Compute "effective per-token weight" for junk vs control under unpacked vs packed	Derived weights/ratios	Reported 8.17× → 1.0× after packing	Supports plausibility of implicit token reweighting mechanism	8.17× derivation unclear given stated mean lengths; depends on true loss reduction and batching

Exp ID	Objective/Hypothesis	Setup (data/split/protocol/baselines)	Metrics	Main Outcome	Claim Supported	Current Limitation
E4	Efficiency measurement of packing	Compare CPT steps/time/tokens/sec across conditions	CPT steps, CPT time, tokens/sec, total time	Huge step reduction and 67× CPT speedup claimed	Packing is a strong efficiency win	Needs normalization by token budget and inclusion of packing overhead; interpretability depends on what is held constant
E5	RULER sub-task recovery analysis	Break down RULER into 13 sub-tasks and report recovery	Per-sub-task recovery and absolute scores	≥95% recovery on 10/13 sub-tasks; Variable Tracking reverses trend (unpacked best)	Packing helps most long-context sub-tasks	Variable Tracking anomaly not diagnosed; need variance and per-task interpretation
E6	Length bucket robustness within junk tweets (packed only)	Partition tweets by length buckets; train packed CPT per bucket	ARC-CoT, RULER	Small variance across buckets	Mitigation effect not sensitive to tweet length within tested range	Does not test semantic-quality independence; length is not a semantic proxy

Research-Theme Gap Diagnosis

- **New knowledge (mechanism)**: Currently *suggestive* but not yet causal. The central hypothesis depends on the actual loss reduction implementation and confound controls (packing mask/composition, step count).
- **Reproducibility/reusability**: Code is linked, but the paper lacks a config table + evaluation protocol details needed for third-party replication.
- **Practice impact**: Potentially high if validated (packing is widely used), but "safe" guidance must be bounded and failure modes (Variable Tracking) must be documented.

Proposed Research Experiments (P0/P1/P2)

Priority	Target Claim	Hypothesis	Minimal Design	Controls/ Baselines	Metrics	Success Criterion	Est. Cost/ Time	Expected Paper-Quality Gain
P0	C1 mechanism (token-weight disparity causes degradation)	Removing length-induced token weighting without packing removes degradation	Re-run **unpacked junk** with **per-token batch-mean loss** (or token-based sampling), matched tokens	Unpacked junk (original), packed junk	ARC-CoT, RULER; optionally "thought-skipping" rate	Unpacked+token-avg approaches packed performance	Low–Med (1 run per seed)	Converts mechanism from correlational to causal
P0	Packing side effects (composition vs weighting)	Cross-document context / masking choices affect results	Compare packed strict block-diagonal mask vs packed naive causal mask	Packed strict vs packed naive; unpacked	ARC-CoT, RULER sub-tasks (esp. Variable Tracking)	Strict vs naive difference is small OR explains anomalies	Low–Med	Clarifies what "packing" means and where it helps/hurts
P0	Step-count confound	Step count differences partly explain drift	Match optimizer updates across packed/unpacked via grad accumulation at matched tokens	Packed vs unpacked matched steps+tokens	ARC-CoT, RULER	Recovery persists when steps are matched	Med	Removes a major alternative explanation
P1	Long-context generality	Mitigation holds beyond 4K	Evaluate RULER at 8K/16K (if model supports) under same checkpoints	No-CPT, packed, unpacked	RULER aggregate + sub-tasks	Trend persists at longer length	Med	Strengthens long-context claim

Priority	Target Claim	Hypothesis	Minimal Design	Controls/ Baselines	Metrics	Success Criterion	Est. Cost/ Time	Expected Paper-Quality Gain
P1	Semantic-quality test	Content quality still matters when weighting is controlled	Fixed packing+loss; vary junk/control mixture ratio	Mixtures 0/25/50/75/100% junk	ARC, RULER, safety/toxicity if relevant	Smooth/weak dependence indicates dynamics dominant; strong dependence indicates semantics matters too	Med	Makes the semantic-vs-dynamics conclusion defensible
P2	Cross-model replication	Not model-specific	Repeat on a second model size/family	Same 3-condition design	ARC, RULER	Similar qualitative pattern	High	Upgrades external validity

ASCII Diagram — Paper Structure & Evidence Map

[Claim: 'Brain Rot' is training artifact from token-weight disparity]
  -> Evidence: Eq.(1)-(3) implies weight ~ 1/L (if per-seq reduction)
  -> Evidence: Table 2 reports 8.17× disparity
  -> Gap: need to verify loss reduction & derive 8.17× transparently

[Claim: Packing mitigates degradation]
  -> Evidence: Table 1 pattern B≈A≫C (ARC, RULER)
  -> Gap: packing also changes steps + context composition
  -> Fix: token-avg loss ablation + strict vs naive packing + matched steps

[Claim: Long-context recovery]
  -> Evidence: RULER sub-task recovery on 10/13
  -> Contradiction: Variable Tracking reversal
  -> Fix: evaluation audit + bounded conclusion

ASCII Diagram — Experiment Upgrade Plan

```
P0 (this week)
  E-Unpacked-TokenAvg  ---> validates/invalidates token-weight mechanism
  E-Packed-Strict-vs-Naive ---> isolates composition/masking effects
  E-StepMatched              ---> removes optimizer-step confound

P1 (next)
  E-RULER-LongerContexts ---> strengthens long-context claim
  E-JunkMixSweep          ---> directly tests semantic-quality dependence

P2 (later)
  E-CrossModelReplication ---> external validity
```

Novelty Verification & Related-Work Matrix

9A) Contribution Novelty Verdict Board

Claim ID	Author Contribution Claim	Key Evidence Papers [n]	Novelty Verdict Tag	Why	Confidence	Required Repositioning
C1	Brain Rot degradation is primarily a **training artifact** caused by **length-induced per-token weight disparity** under per-sequence loss reduction.	[1][6]	`partially_overlapping`	Token-level loss weighting/dynamics are an established lever in LM training (explicitly studied in [6]); [1] documents Brain Rot but does not (in retrieved evidence) attribute it to this specific mechanism. The novelty is mainly the *hypothesis + application* to Brain Rot, but it is not yet causally isolated from confounds (loss reduction implementation, step-count, sequence composition).	Medium	Reframe as: "We hypothesize and provide evidence consistent with an implicit token-weighting contributor to Brain Rot under a common CPT objective; we add controls to isolate it."

Claim ID	Author Contribution Claim	Key Evidence Papers [n]	Novelty Verdict Tag	Why	Confidence	Required Repositioning
C2	**Token-balanced packing** (packing short sequences into fixed-length blocks) eliminates the token-weight disparity ($8.17\times \rightarrow 1.0\times$) and preserves capability.	[2][4][5]	`substantially_overlapped`	Packing with masking to avoid cross-contamination is well-established for efficiency/correctness ([2], practical implementations in [4]); packing strategies that consider sample relatedness exist ([5]). The method described in the manuscript largely matches standard packing; novelty is not in the packing algorithm itself.	High	Drop novelty language about packing as a new technique; reposition as "packing as a best-practice mitigation for a specific CPT objective pathology," with explicit masking/reduction details.

Claim ID	Author Contribution Claim	Key Evidence Papers [n]	Novelty Verdict Tag	Why	Confidence	Required Repositioning
C3	Controlled experiments show packing yields near-full **ARC/RULER recovery** and large **training speedups**, implying short-text CPT is "safe" with packing.	[1][3]	unclear	The empirical pattern is interesting, but decisive novelty vs the Brain Rot protocol depends on whether [1] already controlled for token-weighting/packing and on whether packing effects are separable from sequence-composition effects highlighted in [3]. Given only abstract-level verification for [1], and missing controls in the current manuscript, novelty and causal interpretation remain uncertain.	Low–Medium	Bound the claim to the tested pipeline (model/domain/context length) and present as "mitigation evidence under this protocol," pending ablations + verification of [1]'s training details.

9B) Related-Work Taxonomy Matrix (Layered)

Taxonomy Layer	Branch/Leaf	Representative Papers [n]	Common Assumptions	Difference vs This Paper	Novelty Risk Signal
Root	CPT degradation & mitigation under variable-length training	—	CPT can drift capabilities; training procedures matter	This paper targets short-sequence CPT degradation and a packing-based mitigation	—

Taxonomy Layer	Branch/Leaf	Representative Papers [n]	Common Assumptions	Difference vs This Paper	Novelty Risk Signal
Branch	Brain Rot / data-quality-driven degradation	—	Degradation attributed to content/social/semantic factors	This paper argues for a procedural/optimization artifact explanation	High claim-conflict risk vs [1] unless controls are strong
Leaf	Junk-data CPT capability degradation (“Brain Rot”)	[1]	Controlled CPT on junk vs control corpora can degrade reasoning/long-context	This paper proposes a different root cause + mitigation	Overlap in setting; mechanism claim needs stronger identification
Branch	Packing for efficiency & correctness	—	Packing improves utilization; correctness depends on masking	This paper treats packing as a *capability-preserving* intervention	Method novelty risk is high
Leaf	Packing without cross-contamination / equivalence framing	[2]	Proper masks can make packed training equivalent to unpacked objective	This paper needs to state whether it uses equivalence-style packing	If method = standard packing, novelty is mainly empirical/application
Leaf	Practical packing + masking in modern toolchains	[4]	Packing efficiency depends on implementation details (masking/position ids)	This paper should specify its implementation to support causal claims	Without details, reviewers may attribute gains to engineering choices
Leaf	Semantically related packing strategies (SFT)	[5]	Packing choices affect learning signal and contamination risk	Suggests composition effects that compete with “token-weight only” story	Highlights an alternative mechanism axis (composition)
Branch	Sequence composition & masking effects	—	Concatenation changes effective context distribution	Packing can help/hurt depending on masking and document relations	Confound risk for causal attribution
Leaf	Intra-document masking / retrieval-based sequence construction	[3]	Naive cross-doc causal masking can distract; related-doc packing helps	This paper must control strict vs naive packing to isolate weighting	If not controlled, sequence composition is a plausible alternative explanation

Taxonomy Layer	Branch/Leaf	Representative Papers [n]	Common Assumptions	Difference vs This Paper	Novelty Risk Signal
Branch	Token-level weighting/dynamics	—	Token weights steer what the model learns	This paper's mechanism is an *implicit* token weighting via length	Overlap with token-weighting literature
Leaf	Explicit token weighting for long-context training	[6]	Non-uniform token weights can improve long-context behavior	This paper's weighting is length-induced and may be unintended	Positions your mechanism as a special case of token-weight steering

9C) Head-to-Head Comparison Matrix

Ref [n]	Problem/Setting	Method Core	Strongest Overlap Point	Clear Difference	Impact on Final Judgment
[1]	CPT on junk social-media text causing capability decline ("Brain Rot")	Controlled junk/control corpora + evaluation; attributes to data quality in retrieved evidence	Same phenomenon + similar benchmarks/pipeline motivation	This paper proposes a different mechanism (implicit token weighting) and mitigation (packing)	Novelty depends on whether [1] already controlled for packing/loss reduction; must verify and align protocols
[2]	Packing variable-length sequences efficiently without cross-contamination	Formal packing + masking to preserve equivalence	Packing as a known technique	This paper's "token-balanced packing" is not algorithmically new vs [2]	Reposition: your novelty is *why packing matters for CPT stability* and empirical validation
[3]	How sequence composition/concatenation affects pretraining generalization	Intra-document masking + retrieval-based construction	Shows packing/concatenation changes context distribution and performance	Competes with "token-weight only" explanation; suggests masking/composition controls are needed	Strengthens the need for strict-vs-naive packing ablation
[4]	Engineering analysis of packing with correct masking (FlashAttention/HF)	Implementation details for masking-aware packing	Highlights that practical packing correctness is non-trivial	This paper currently omits those implementation specifics	Without specifying masking/position ids, the mechanism claim is under-supported

Ref [n]	Problem/Setting	Method Core	Strongest Overlap Point	Clear Difference	Impact on Final Judgment
[5]	Packing for SFT with sample-relatedness selection	Select related samples within packs to improve learning and reduce contamination	Emphasizes semantic composition within packs	This paper focuses on CPT and implicit weighting, but [5] suggests composition can matter	Encourages testing whether related-sample packing changes results
[6]	Improving long-context LMs via explicit token weighting	Token-weighting schemes based on dependency/uncertainty	Confirms that token weighting matters for long-context behavior	Different goal (explicit steering) but overlaps in "loss weights matter" thesis	Supports "partially overlapping" novelty for the token-weight mechanism

ASCII Diagram — Related-Work Taxonomy Tree (Layered)

```
Related Work Taxonomy (Root: CPT degradation & mitigation under variable-length
training)
 ├── Branch 1: Brain Rot / data-quality-driven degradation
 │    └── Leaf 1.1: Junk-data CPT capability degradation studies [1]
 ├── Branch 2: Packing for efficiency & correctness
 │    ├── Leaf 2.1: Packing without cross-contamination / equivalence [2]
 │    ├── Leaf 2.2: Practical packing implementations & masking details [4]
 │    └── Leaf 2.3: Packing strategies using sample relatedness (SFT) [5]
 ├── Branch 3: Sequence composition & masking effects
 │    └── Leaf 3.1: Intra-document masking / retrieval-based sequence construction [
3]
 └── Branch 4: Token-level weighting/dynamics
     └── Leaf 4.1: Explicit token weighting for long-context training [6]

This paper's intended position:
 - Uses packing (Branch 2) as a mitigation, but argues the key value is
diagnosing an
   implicit token-weight pathology (Branch 4) as a contributor to Brain Rot (
Branch 1).
 - Must rule out Branch 3 (composition/masking) as the primary explanation via
controls.
```

Contribution-level Novelty Conclusion: C1 is best viewed as **partially overlapping** with existing token-weighting/dynamics work ([6]) but potentially novel in its *application to Brain Rot* ([1])—provided the manuscript verifies the loss reduction implementation and adds confound-controlling ablations. C2 is **substantially overlapped** because packing itself is established ([2][4][5]); the paper should reposition away from "new packing" and toward "packing as a mitigation + diagnosis under a specific CPT objective pathology." C3 remains **unclear** until the Brain Rot protocol details in [1] and sequence-composition alternatives ([3]) are more fully controlled/compared.

References

[1] LLMs Can Get "Brain Rot"! 2510.13928

[2] Efficient Sequence Packing without Cross-contamination: Accelerating Large Language Models without Impacting Performance 2107.02027

[3] Analysing The Impact of Sequence Composition on Language Model Pre-Training 2402.13991

[4] Enhancing Training Efficiency Using Packing with Flash Attention 2407.09105

[5] Threshold Filtering Packing for Supervised Fine-Tuning: Training Related Samples within Packs 2408.09327

[6] Token Weighting for Long-Range Language Modeling 2503.09202

Scores

Final Score: 6.0/10

Post-Revision Target: [7.5, 8.5]/10

Score rationale (novelty + research value prioritized): The work has strong practical value if validated (packing is widely used and the reported mitigation effect is large), but the current manuscript has major validity/defensibility gaps that directly affect the central novelty claim: loss reduction implementation is unspecified, the 8.17× statistic is not transparently derived, packing confounds are not isolated, and long-context evidence includes an unresolved contradictory sub-task. With P0 fixes (objective verification + targeted ablations + bounded claims + reproducibility tables), the contribution could become a solid, practice-shaping note on how training procedure choices can induce or mitigate Brain Rot-like degradation.

Appendix · Original Paper PDF

The following pages preserve source layout and include review annotations when available.

The appendix below attaches source paper pages and overlays review highlights for ISSUE / SUGGESTION / EVIDENCE objects.

Callout boxes are placed near page margins to avoid covering core content and remain within page boundaries.

End of report · DeepScientist Review Export

TOKEN-BALANCED CONTINUAL PRETRAINING ELIMINATES BRAIN ROT DEGRADATION

FARS
Analemma
fars@analemma.ai

ABSTRACT

Continual pretraining (CPT) on low-quality, short-sequence social media posts has been shown to cause "Brain Rot"—severe degradation in reasoning and long-context capabilities. The prevailing explanation attributes this to the semantic quality of the data itself. We challenge this assumption and demonstrate that Brain Rot is a *training artifact* arising from per-token weight disparity: when short sequences are processed without packing, they receive 8.17× higher per-token gradient updates than longer sequences, causing their statistical patterns. We propose token-balanced packing, which concatenates sequences to uniform length, eliminating this disparity. Across 4 experiments on Llama-3-8B-Instruct, we show that packing recovers 98.7% reasoning recovery and 95.2% RULER long-context recovery relative to the no-CPT baseline, while providing a 67× training speedup. Our findings demonstrate that CPT on short-sequence data is safe when proper packing is employed.

WARNING: This paper was generated by an automated research system. The code is publicly available.[1]

Index ID: #P01-A06
Full text moved to continuation pages.

#P01-A01 · ISSUE · MAJOR · #R001 · INDEX · CONT.
Index ID: #P01-A01
Full text moved to continuation pages.

1 INTRODUCTION

Continual pretraining (CPT) has emerged as a powerful paradigm for adapting large language models to specialized domains (Gururangan et al., 2020; Ke et al., 2023). By continuing the pretraining process on domain-specific corpora, practitioners can efficiently inject new knowledge without the computational cost of training from scratch. However, recent work has identified a concerning phenomenon termed "Brain Rot" (Xing et al., 2025), where CPT on low-quality, short-sequence data such as social media posts leads to severe degradation in both reasoning and long-context understanding.

Index ID: #P01-A02
Full text moved to continuation pages.

The prevailing explanation for Brain Rot attributes the degradation to the semantic quality of the training data itself—the assumption being that exposure to informal, fragmented text corrupts the model's learned representations. This interpretation has led to recommendations for aggressive data filtering and avoidance of short-sequence corpora in CPT pipelines. However, we hypothesize that Brain Rot is not an inherent property of low-quality content, but rather a *training artifact* arising from how standard training procedures handle variable-length sequences.

Index ID: #P01-A03
Full text moved to continuation pages.

In this paper, we investigate the root cause of Brain Rot through the lens of gradient dynamics. We observe that when training on short sequences without packing, the loss is averaged equally to the batch loss regardless of length, resulting in shorter sequences receiving disproportionately high per-token gradient updates. This per-token weight disparity causes the model to overfit to the statistical patterns of short sequences. We propose *token-balanced packing*, which concatenates multiple short sequences to form uniform-length training examples, thereby equalizing per-token contributions and eliminating the disparity.

Index ID: #P01-A04
Full text moved to continuation pages.

#P01-A05 · ISSUE · MAJOR · #R005 · INDEX · CONT.
Index ID: #P01-A05
Full text moved to continuation pages.

Our contributions are as follows:

- We demonstrate that Brain Rot is a training artifact caused by per-token weight disparity (8.17× in our experiments), not an inherent property of low-quality semantic content.

[1] https://gitlab.com/fars-a/token-balanced-cpt-brain-rot

Annotation Continuation · Source Page 1

Continuation sheet linked to source highlights. Click marker headers to jump back to the paper region.

#P01-A01 · ISSUE · MAJOR · #R001 · Part 1

Page 1 - Abstract

The abstract currently mixes (i) a *mechanistic claim* (“Brain Rot is a training artifact from per-token weight disparity”) with (ii) a *quantitative recovery claim* (“119.7%/95.2% recovery relative to the no-CPT baseline”) and (iii) a *broad safety claim* (“CPT on short-sequence data is safe”). In the current manuscript, the recovery metric is later defined using Conditions A/B/C (not the no-CPT baseline), and the packed conditions still underperform the no-CPT baseline on RULER. As written, the abstract risks misleading readers about what was actually recovered.

Why it matters: the abstract is what most readers will quote. If the recovery definition and “safe” scope are not consistent with the reported tables, reviewers will treat this as a reliability red flag even if the experiments are otherwise solid.

Actionable fix:
· Replace “relative to the no-CPT baseline” with the exact recovery definition used in the experiments (or change the metric to be baseline-relative and recompute).
· Bound “safe” to the tested setup (model, dataset, and evaluation), and make clear that packing prevents *additional degradation relative to unpacked CPT* rather than guaranteeing no regression vs. no-CPT.

Mentor Revised Version:
Continual pretraining (CPT) on short, low-quality text has been reported to degrade reasoning and long-context performance (“Brain Rot”). We test whether this degradation can arise from training dynamics when variable-length sequences are optimized with per-sequence-averaged loss, which upweights short samples at the token level. We apply token-balanced packing that concatenates short sequences into fixed-length blocks, equalizing per-token contributions under the same loss reduction. In controlled CPT+instruction-tuning experiments on Llama-3-8B-Instruct using the Brain Rot datasets, packing largely closes the performance gap between packed and unpacked CPT on ARC-Challenge and RULER, while substantially reducing the number of training steps. These results suggest that capability degradation from short-sequence CPT can be mitigated by token-balanced training, though broader safety claims require validation across models, corpora, and packing/masking implementations.

#P01-A02 · SUGGESTION · MINOR · #R002 · Part 1

Page 1 - Abstract / Disclaimer

Including an explicit disclaimer about automated generation is unusually transparent and helpful. However, as written it does not tell a reviewer what parts were automatically generated vs. manually verified, nor what checks were performed to ensure experimental correctness.

Why it matters: for an empirical paper making a strong causal claim (“training artifact”), credibility depends on reproducibility and on demonstrating that the pipeline (data, packing/masking, training) was audited. Without that, the disclaimer can backfire by priming skepticism.

Actionable fix:
· Add 2–3 sentences stating what was automatically generated (e.g., writing/plots) and what was validated by humans (e.g., training runs, dataset construction, evaluation scripts).
· In the code link/appendix, add a “Repro checklist” (exact commit hash, dataset hashes, training command lines, seed control, packing/masking implementation details, and evaluation prompts).

Mentor Revised Version:
WARNING: This manuscript was drafted by an automated research system. All experiments, dataset preprocessing scripts, and evaluation pipelines were manually verified and are reproducible from the released code (commit hash: XXXX) and dataset manifests (checksums: XXXX). We provide exact training commands, seeds, packing/masking implementation details, and evaluation prompts to enable independent replication.

#P01-A03 · SUGGESTION · MAJOR · #R003 · Part 1 · CONT.

Page 1 - Introduction (opening paragraph)

This opening paragraph sets context for continual pretraining, but it introduces “Brain Rot” only as a name + qualitative degradation. Given that your paper’s core contribution is a mechanistic explanation for that phenomenon, the introduction should define it in operational terms early (which tasks/metrics drop, under what CPT pipeline, starting from what checkpoint).

Why it matters: without an operational definition, readers cannot tell whether your study matches the prior setup you cite or whether “Brain

Annotation Continuation · Source Page 1 (cont. 2)

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#P01-A03 · SUGGESTION · MAJOR · #R003 · Part 2

Rot" is being redefined. This also weakens later causal language because the target effect is not crisply specified.

Actionable fix:
· Add one sentence that pins down the *evaluation signals* used to quantify Brain Rot (e.g., ARC-Challenge CoT and RULER) and the *pipeline* (CPT → instruction tuning) used in both prior work and your replication.
· If you start from an instruction-tuned checkpoint (as later stated), mention that here, because it changes interpretation versus starting from a base pretrained model.

Mentor Revised Version:
Continual pretraining (CPT) is widely used to adapt large language models to new domains by continuing next-token training on domain corpora. Recent work reports that CPT on short, low-quality social media text can substantially reduce measured reasoning and long-context performance (e.g., drops on ARC-Challenge with chain-of-thought prompting and on RULER-style long-context tasks) [1]. In this paper we revisit that setting under a controlled CPT→instruction-tuning pipeline and ask whether the observed degradation can be explained by training dynamics specific to variable-length short sequences.

References:
[1] LLMs Can Get "Brain Rot"! 2510.13928

#P01-A04 · SUGGESTION · MAJOR · #R004 · Part 1

Page 1 - Introduction (prevailing explanation paragraph)

This paragraph sets up the contrast between "semantic quality" vs "training artifact," but it stays at the level of narrative framing. Given that the paper later uses a 3-condition design to adjudicate hypotheses, the introduction should preview that falsifiable test and acknowledge alternative training-dynamics explanations beyond per-token weighting (e.g., step-count differences, sequence composition/concatenation effects).

Why it matters: reviewers will ask whether the paper truly isolates "semantic quality" from "training dynamics," especially because packing changes more than token weights (it also changes context composition and the number of optimizer updates needed for a fixed token budget). Making the falsification criteria explicit increases defensibility.

Actionable fix:
· Add one sentence that states the decisive prediction: if content drives degradation, then packed junk should still degrade; if token-weight disparity drives it, packed junk should match packed control.
· Add one sentence acknowledging other dynamics that must be controlled (e.g., token budget vs optimizer steps; packing masks / cross-document attention), and point to where you control/measure them.

Mentor Revised Version:
Prior work largely attributes Brain Rot to the semantic properties of low-quality short text, motivating heavy filtering or avoidance of such corpora. Our alternative hypothesis is that a substantial part of the observed degradation can be induced by training dynamics when short samples are optimized under per-sequence-averaged loss, which implicitly upweights short sequences at the token level. This hypothesis makes a falsifiable prediction: if semantic content is the dominant driver, then CPT on packed junk text should still degrade compared to packed control text; if token-weight disparity is dominant, then packed junk and packed control should perform similarly while unpacked junk degrades. Because packing can also change sequence composition and the number of optimizer updates for a fixed token budget, we explicitly control/report these factors in the experimental setup.

#P01-A05 · ISSUE · MAJOR · #R005 · Part 1 · CONT.

Page 1 - Introduction (mechanism preview paragraph)

This paragraph is the paper's causal core: "short sequences without packing ⇒ equal per-sequence loss contribution ⇒ higher per-token gradients ⇒ overfit short patterns." The mechanism is plausible, but as written it hides two implementation-critical assumptions:

(1) The loss reduction is truly per-sequence-averaged (each sequence contributes equally), not per-token over the batch. Many training stacks use token-level averaging (or token-based sampling) by default.

(2) "Without packing" also changes the number of optimizer updates required to process the same token budget. Your later efficiency table

Annotation Continuation · Source Page 1 (cont. 3)

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#P01-A05 · ISSUE · MAJOR · #R005 · Part 2

suggests orders-of-magnitude fewer steps with packing. Step-count differences can themselves change drift/forgetting dynamics, so you need to show the effect persists under a fixed *token budget* and to report both token and step budgets per condition.

Why it matters: if either assumption is violated, the per-token-weight argument is not the unique explanation, which weakens the claim that Brain Rot is "a training artifact" specifically caused by this disparity.

Actionable fix:
· In Methods/Setup, explicitly state the exact loss reduction used (per-sequence mean vs per-token mean) and show a short code snippet or formula matching your training framework.
· Add a control or at least a reporting table: tokens processed, optimizer steps, effective batch size, and learning rate schedule per condition.
· Add a robustness check: re-run unpacked with token-level loss reduction (or token-based sampling) to see if degradation disappears without packing.

Mentor Revised Version:
We hypothesize that the reported degradation can be induced by a mismatch between variable-length data and the loss reduction used during CPT. Specifically, when the training objective averages cross-entropy per sequence and then averages equally across sequences, tokens from shorter sequences receive larger effective gradient weight than tokens from longer sequences. In contrast, packing short sequences into fixed-length blocks (with correct attention masking) makes per-token contributions approximately uniform under the same reduction. To ensure this mechanism is the relevant causal factor, we explicitly report (and where possible control) the loss reduction implementation, the total token budget, and the number of optimizer updates for each condition.

#P01-A06 · SUGGESTION · MAJOR · #R019 · Part 1

Page 1 - Introduction: Contributions list

The contributions section is currently truncated in the PDF extract (only the first bullet is visible), and even in the manuscript markdown it mixes mechanism, results, and efficiency claims. This is a missed opportunity: a clean contributions list is the main "contract" with reviewers.

Why it matters: if contributions are not crisply stated with scope bounds and evidence pointers, reviewers will interpret later strong claims (e.g., "safe") as overreach.

Actionable fix:
· Rewrite as exactly 3 contributions (C1–C3), each with: (i) what is claimed, (ii) what is *new*, (iii) what evidence section supports it, and (iv) scope limits.
· Avoid framing performance numbers as a standalone contribution; attach numbers as evidence under the methodological contribution.

Mentor Revised Version:
Our contributions are threefold. (1) We identify a training-dynamics mechanism that can induce Brain Rot-like degradation under a common CPT objective: when loss is reduced per sequence, variable-length short samples receive disproportionately large effective per-token gradient weight (Methods: Sec. 3.1–3.2). (2) We propose token-balanced packing with correct masking to equalize per-token contribution while preserving the intended training objective, and we describe the implementation details needed to avoid cross-sample attention (Methods: Sec. 3.2). (3) In controlled CPT→instruction-tuning experiments on Llama-3-8B-Instruct using the Brain Rot datasets, we show that packed CPT substantially mitigates the degradation seen in unpacked CPT on ARC-Challenge and RULER, while reducing the number of optimizer steps and wall-clock time; we also document remaining failure modes (Experiments: Sec. 4–5).

- We show that token-balanced packing achieves full capability recovery, with 119.7% ARC reasoning recovery and 95.2% RULER long-context recovery relative to the no-CPT baseline.
- We provide mechanistic analysis showing that packing eliminates per-token weight disparity by equalizing gradient contributions across sequences.
- We demonstrate that packing provides a 67× training speedup compared to unpacked training, making it both more effective and more efficient.

2 Related Work

Continual Pretraining. Continual pretraining (CPT) adapts pretrained language models to specific domains by further training on domain-specific corpora (Gururangan et al., 2020). This approach has proven effective for adapting models to specialized domains such as biomedical text, legal documents, and scientific literature. Ke et al. (2023) systematically studied CPT strategies and identified key factors affecting knowledge retention and acquisition. Scaling laws (Kaplan et al., 2020; Hoffmann et al., 2022) provide theoretical foundations for understanding how model and data size affect performance, though these analyses typically assume well-curated, long-form documents rather than short, informal text.

Data Quality in Pretraining. Data quality significantly impacts language model performance. Wenzek et al. (2019) developed CCNet for extracting high-quality monolingual data from web crawls using perplexity filtering. Deduplication has emerged as a critical preprocessing step, with both exact (Lee et al., 2021) and semantic (Abbas et al., 2023) approaches improving training efficiency and model quality. Wettig et al. (2024) proposed QuRating for selecting high-quality training data based on model-derived quality signals. Most relevant to our work, Xing et al. (2025) identified the "Brain Rot" phenomenon where CPT on short social media posts severely degrades model capabilities, attributing this to low-quality semantic content. Our work challenges this attribution by demonstrating that the degradation stems from training dynamics rather than content quality.

Sequence Length and Training Efficiency. Variable sequence [illegible] that have motivated several optimization strategies. Jin et al. (2023) proposed progressively growing training length to accelerate pretraining, while Pouransari et al. (2024) introduced dataset decomposition with variable-length curricula. Sequence packing (Krell et al., 2022) concatenates multiple sequences to fill fixed-length blocks, improving GPU utilization. Li et al. (2024) addressed workload imbalances in distributed training over variable-length sequences. Helm et al. (2025) explored token weighting strategies for long-range language modeling. Our work reveals that packing not only improves efficiency but also eliminates capability degradation from short-sequence CPT by equalizing per-token gradient weights.

Long-Context Evaluation. Evaluating long-context capabilities requires specialized benchmarks. Liu et al. (2023) demonstrated that language models struggle to use information in the middle of long contexts. RULER (Hsieh et al., 2024) provides a comprehensive benchmark with multiple sub-tasks including retrieval, aggregation, and question answering at various context lengths. We use RULER alongside ARC-Challenge (Clark et al., 2018) with chain-of-thought prompting (Wei et al., 2022) to evaluate both long-context and reasoning capabilities.

3 Method

3.1 Problem Formulation

Consider continual pretraining (CPT) on a dataset $\mathcal{D} = \{x_1, x_2, \ldots, x_N\}$ where each sample x_i is a sequence of L_i tokens. Under standard cross-entropy loss with per-sample averaging, the loss for sample x_i is:

$$\mathcal{L}_i = -\frac{1}{L_i} \sum_{t=1}^{L_i} \log p_\theta(x_i^{(t)} | x_i^{(<t)})$$

Index ID: #P02-A01
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#P02-A02 · ISSUE · MAJOR · #R007 · INDEX · CONT.
Index ID: #P02-A02
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#P02-A01 · SUGGESTION · MAJOR · #R006 · Part 1

Page 2 - Related Work

This section provides a reasonable high-level landscape, but it currently (i) stays mostly at textbook-level background and (ii) mixes your paper's claim ("packing … eliminates capability degradation … by equalizing per-token gradient weights") into the related-work narrative without clearly separating what prior work established vs what you newly demonstrate. It also misses several "nearest neighbor" strands that matter directly for your causal story:

- **Packing correctness and cross-contamination**: packing is not just an efficiency trick; correctness depends on attention masking/position handling. Prior work explicitly formalizes packing and discusses equivalence / cross-contamination avoidance [2] and more recent engineering analyses emphasize masking details in modern stacks [4].
- **Sequence composition effects**: concatenation/packing can change the effective context distribution; there is evidence that naive causal masking across document boundaries can inject distracting context, while intra-document masking or related-document packing can improve generalization [3]. This is directly relevant because your argument treats packing primarily as a gradient-weight fix.
- **Token-level weighting / token-level dynamics**: there is adjacent work on deliberately weighting tokens to improve long-context learning [6]. Even if your mechanism differs (length-induced implicit weighting), readers will expect you to position against "explicit token weighting" as a sibling solution.

Why it matters: without these comparisons, novelty is harder to judge and the mechanistic claim looks under-controlled (packing changes more than per-token weights). A sharper related-work matrix will also help you bound the claim "Brain Rot is not semantic" to the specific training dynamics you test.

Actionable fix:
1) Reorganize related work by *decision-relevant axes* (data-quality hypothesis vs training-dynamics hypothesis; packing correctness; sequence composition; token-weighting).
2) Add 1–2 sentences per axis stating the concrete gap and how your method differs.
3) Remove/relocate self-claims from related work into Introduction/Method ("we show …") so the section remains about prior literature.

Mentor Revised Version:
Prior work reports severe capability degradation during continual pretraining on short, low-quality social media text ("Brain Rot") and largely attributes it to data quality and social/semantic factors [1]. In parallel, the training community has developed sequence packing to reduce padding and improve throughput, with formal treatments emphasizing attention masking to avoid cross-contamination and preserve mathematical equivalence to unpacked training [2], as well as practical implementations in modern toolchains [4]. However, packing and document concatenation also change the context distribution seen during training; recent analyses show that naive cross-document causal masking can introduce distracting context, and that intra-document masking or constructing packs from related documents can improve generalization [3]. Finally, token-level loss weighting has been explored as an explicit mechanism to steer long-context learning [6]. Our work connects these strands by testing whether a length-induced, *implicit* token-weight imbalance in standard per-sequence loss reduction can itself induce Brain Rot-like degradation, and whether token-balanced packing can mitigate it under controlled CPT settings.

References:
[1] LLMs Can Get "Brain Rot"! 2510.13928

[2] Efficient Sequence Packing without Cross-contamination: Accelerating Large Language Models without Impacting Performance 2107.02027

[3] Analysing The Impact of Sequence Composition on Language Model Pre-Training 2402.13991

[4] Enhancing Training Efficiency Using Packing with Flash Attention 2407.09105

[6] Token Weighting for Long-Range Language Modeling 2503.09202

#P02-A02 · ISSUE · MAJOR · #R007 · Part 1 · CONT.

Page 2 - Method: Problem Formulation (Eq. 1)

Eq. (1) defines a per-sequence average cross-entropy, which is fine mathematically. The key problem is that your central mechanism ("short sequences receive higher per-token gradient weight") only holds if your *training implementation* computes (i) a per-token mean within each sequence and then (ii) an equal-weight mean across sequences (as in your Eq. (2) on the next page). Many LLM training stacks instead compute a token-level mean across the entire batch (i.e., divide by the number of non-masked tokens), which would largely remove the length-induced gradient disparity *even without packing*.

Annotation Continuation · Source Page 2 (cont. 2)

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#P02-A02 · ISSUE · MAJOR · #R007 · Part 2

Why it matters: if the loss reduction in your actual CPT runs is token-averaged, then the per-token-weight explanation is not the causal driver, and packing may be helping via other mechanisms (sequence composition, context distribution, or step-count effects). This is validity-critical because it underpins the paper’ s headline claim.

Actionable fix:

· Add an explicit “Implementation of loss reduction” paragraph + 5–10 lines of pseudocode for how you compute loss given an attention mask.
· Report whether you average loss per sequence, per token, or per packed block, and how padding/EOS tokens are handled.
· Add a control: run the ***unpacked*** condition with a token-averaged reduction (or token-based sampling) and check whether Brain Rot degradation disappears without packing.

Mentor Revised Version:

We consider next-token CPT with attention masking and explicitly distinguish two common loss reductions. Given a batch B with sequences x_i and non-masked token counts L_i, the per-sequence reduction is

$$mathcal{L}_{text{seq}}=frac{1}{|B|}sum_{iin B}frac{1}{L_i}sum_{t=1}^{L_i}-log p_theta(x_i^{(t)}mid x_i^{(<t)}),$$

which upweights tokens from shorter sequences by a factor proportional to $1/L_i$. In contrast, the per-token reduction is

$$mathcal{L}_{text{tok}}=frac{1}{sum_{iin B} L_i}sum_{iin B}sum_{t=1}^{L_i}-log p_theta(x_i^{(t)}mid x_i^{(<t)}),$$

which assigns approximately uniform weight per token. In our experiments we use $mathcal{L}_{text{seq}}$ (verify in code) to match the Brain Rot protocol; we additionally report a $mathcal{L}_{text{tok}}$ control to isolate the effect of length-induced token-weight imbalance.

where p_θ denotes the language model parameterized by θ. The total training loss averages over all samples:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^{N} \mathcal{L}_i \tag{2}$$

This formulation implies that each token in sample x_i receives an effective gradient weight of $w_i = \frac{1}{N \cdot L_i}$. Critically, shorter sequences receive higher per-token weights. For two samples with lengths L_{short} and L_{long} where $L_{\text{short}} \ll L_{\text{long}}$, the per-token weight ratio is:

$$\frac{w_{\text{short}}}{w_{\text{long}}} = \frac{L_{\text{long}}}{L_{\text{short}}} \tag{3}$$

When CPT data contains very short sequences (e.g., social media posts with mean length 16.7 tokens) mixed with longer documents (mean length 100.4 tokens), this creates an 8.17× per-token weight disparity. The model disproportionately optimizes for short-sequence patterns, potentially degrading capabilities learned during pretraining.

3.2 Token-Balanced Packing

To eliminate the per-token weight disparity, we apply sequence packing (Krell et al., 2022). Rather than processing each short sequence as an independent sample, we concatenate multiple sequences into fixed-length blocks of L_{pack} tokens (e.g., 2048), separated by end-of-sequence tokens. Figure 1 illustrates this approach.

Under packing, each token receives a uniform gradient weight of $w = \frac{1}{M \cdot L_{\text{pack}}}$, where M is the number of packed blocks. This eliminates the length-dependent weight disparity:

$$\frac{w_{\text{short}}}{w_{\text{long}}} = 1.0$$

Packing provides two key benefits: (1) it equalizes per-token gradient contribution regardless of sequence length, and (2) it dramatically improves training efficiency by reducing the number of forward-backward passes required to process the same number of tokens.

3.3 Experimental Design

To disentangle the effects of semantic content quality from training dynamics, we design a controlled experiment with three conditions that vary two factors: content quality (junk vs. control) and packing (packed vs. unpacked).

Condition A (Control, Packed). CPT on high-quality control documents using packing. This serves as the upper-bound reference for expected performance.

Condition B (Junk, Packed). CPT on low-quality "junk" tweets using packing. If semantic content drives degradation, this condition should perform poorly despite packing.

Condition C (Junk, Unpacked). CPT on junk tweets without packing. This reproduces the Brain Rot phenomenon observed in prior work (Xing et al., 2025).

This design enables hypothesis testing: if semantic content quality is the primary driver of Brain Rot degradation, we expect $A \gg B \approx C$. However, if the degradation is a training artifact from per-token weight disparity, we expect $A \approx B \gg C$. The latter would indicate that packing eliminates the degradation regardless of content quality.

3.4 Evaluation

We evaluate on two complementary benchmarks. ARC-Challenge of-thought prompting (Wei et al., 2022) measures scientific reasoning ability, requiring multi-step

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Page 3 - Method: Per-token weight disparity (Eq. 2–3 + 8.17× example)

Eq. (2–3) correctly show that under a ***per-sequence-averaged*** objective, token weights scale like $1/L_i$. However, the concrete numeric example is currently inconsistent: the paragraph states mean lengths 16.7 and 100.4 tokens and then claims this "creates an 8.17× disparity." Using those arithmetic means, $100.4/16.7approx 6.0$, not 8.17.

A likely resolution is that the effective per-token weight is proportional to $mathbb{E}[1/L]$ (i.e., the **harmonic mean** length), not $1/mathbb{E}[L]$. If so, the paper should say that explicitly and show the computation that produces 8.17× from the actual length distributions.

Why it matters: 8.17× is repeated as a headline figure and is used to motivate the causal mechanism. If the computation basis is unclear, reviewers will doubt the mechanistic grounding even if the qualitative trend holds.

Actionable fix:
· Replace the mean-length ratio narrative with a short derivation using $mathbb{E}[1/L]$ and report both arithmetic mean and harmonic mean length per dataset.
· Clarify what N denotes in practice (dataset size vs batch size vs number of sequences per optimizer step), especially if you use token-based dynamic batching where N varies with length.

Mentor Revised Version:
Under the per-sequence reduction in Eq. (2), each sequence contributes weight $1/N$ to the batch loss, and each non-masked token within sequence i therefore receives effective weight proportional to $1/L_i$. For a dataset or batch with a length distribution, the relevant aggregate quantity is $mathbb{E}[1/L]$ (equivalently, the harmonic mean length), not $1/mathbb{E}[L]$. In our CPT data, the junk-tweet length distribution yields $mathbb{E}[1/L]_{text{junk}}$ and the control-document distribution yields $mathbb{E}[1/L]_{text{ctrl}}$, giving a per-token weight ratio of $mathbb{E}[1/L]_{text{junk}}/mathbb{E}[1/L]_{text{ctrl}}approx 8.17$. We report both arithmetic and harmonic mean lengths and the exact computation used to obtain this ratio.

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Page 3 - Method: Token-balanced packing (Eq. 4 + paragraph)

This packing subsection correctly states the high-level idea, but it omits the technical detail that determines whether packing is "mathematically equivalent" to unpacked training: the attention mask and position/segment handling must prevent cross-sequence contamination (unless you intentionally allow it). This matters for your causal story, because packing can also *change the learned distribution* by exposing tokens to cross-document context under standard causal masking, which has been shown to hurt or help depending on construction [3].

Why it matters: if packing is implemented with imperfect masking (or with cross-document attention), then the observed recovery could be partly due to altered context composition rather than purely due to token-weight equalization. Reviewers will ask for this.

Actionable fix:
· Add a 4–6 line "Packing correctness" paragraph describing the exact implementation: (i) whether attention is blocked across packed sequences, (ii) whether position ids reset per sample, and (iii) how EOS separators are treated.
· State explicitly whether you aim for equivalence to unpacked training (as in [2]) or whether you allow cross-document context.
· Add a short note acknowledging sequence composition effects and why you believe they are not the primary driver here (or add a control: packed with cross-attention allowed vs packed with strict masking).

Mentor Revised Version:
We implement token-balanced packing by concatenating multiple short sequences into a fixed-length block and inserting an EOS separator between sequences. To avoid cross-contamination, we apply a block-diagonal causal attention mask so that tokens can only attend to earlier tokens within the same original sequence, and we reset position ids at the start of each sequence (equivalently, we use per-sequence position offsets). This makes the packed objective mathematically equivalent to unpacked per-sequence training while equalizing the per-token weighting induced by Eq. (2) [2]. Because document concatenation strategies can also change the effective context distribution and introduce distracting cross-document context [3], we additionally report a control comparing strict-masked packing versus naive causal masking across the pack.

References:
[2] Efficient Sequence Packing without Cross-contamination: Accelerating Large Language Models without Impacting Performance 2107.02027

[3] Analysing The Impact of Sequence Composition on Language Model Pre-Training 2402.13991

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Page 3 - Method: Experimental Design (Conditions A/B/C)

The A/B/C design is a strong start for separating "junk content" from "packing," but it still leaves two important confounds relative to your mechanistic claim "per-token weight disparity is the root cause":

1) **Optimizer updates / learning dynamics**: packed training uses far fewer steps for a fixed token budget. If you keep epochs fixed, token budget differs; if you keep tokens fixed, step count differs. Either can affect drift/forgetting. The design needs an explicit statement of what is held constant (tokens, steps, or wall-clock) and at least one control that rules out step-count as the primary driver.

2) **Sequence composition effects**: packing changes context composition; naive causal masking can introduce cross-document context, which has independent effects on generalization [3]. Without specifying strict masking/equivalence, "packing vs unpacking" is not a single-factor intervention.

Why it matters: absent these controls, the safest scientific claim is "packing mitigates the degradation under this protocol," not "the root cause is per-token weight disparity." If the paper keeps the stronger claim, it needs stronger identification.

Actionable fix:
· Add a small "What we control" table: tokens processed, optimizer steps, sequence composition/masking mode, LR schedule, batch size.
· Add at least one additional condition or ablation:
(i) Unpacked + token-averaged loss reduction (tests whether weighting alone explains the effect), and/or
(ii) Packed with strict block-diagonal attention vs packed with naive causal attention (tests composition effect), and/or
(iii) Match optimizer steps across packed/unpacked by adjusting batch size or gradient accumulation.
· If not feasible, rephrase the conclusion as evidence "consistent with" the weight-disparity hypothesis and explicitly list remaining confounds.

Mentor Revised Version:
We evaluate three primary conditions that vary (a) content quality and (b) whether short sequences are packed into fixed-length blocks. To attribute differences specifically to length-induced token-weight imbalance, we additionally report per-condition token budgets and optimizer-step counts, and we include controls that (i) replace per-sequence loss reduction with per-token reduction in the unpacked setting and (ii) compare strict-masked packing versus naive cross-document causal masking. These controls help separate token-weighting effects from step-count and sequence-composition effects.

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Page 3 - Method: Evaluation

The evaluation paragraph names ARC-CoT and RULER@4K, but it does not provide the minimum details needed to reproduce the reported numbers or to interpret them as evidence for "reasoning" and "long-context" preservation. In particular, ARC-CoT accuracy can be sensitive to prompt format and decoding settings, and RULER contains multiple tasks/length regimes where performance can vary sharply.

Why it matters: your central claim is about *capability degradation/preservation*, so any evaluation degrees of freedom (prompting, decoding, length, scoring) become potential confounders. Reviewers will ask whether the gains are robust to reasonable evaluation variants.

Actionable fix:
· Add an "Evaluation protocol" paragraph (or appendix) that specifies:
(i) exact prompts (including whether you force CoT, allow refusal, or strip reasoning),
(ii) decoding parameters (temperature, top-p/k, max tokens),
(iii) scoring procedure (exact-match vs normalized; how invalid formats are handled),
(iv) for RULER: which sub-tasks are included and whether context length is fixed to 4K or varied.
· Justify why 4K context length is the right test for your "long-context recovery" claim, or add at least one longer-length evaluation point (e.g., 8K/16K) if the base model supports it.

Mentor Revised Version:
We evaluate (a) scientific reasoning using ARC-Challenge under a fixed chain-of-thought prompt and deterministic decoding, and (b) long-context behavior using the RULER benchmark with context length fixed at 4K tokens across all sub-tasks. For each benchmark, we release the exact prompts, decoding settings, and scoring scripts (including how malformed outputs are handled) to ensure results are reproducible. To assess whether conclusions depend on the chosen context length, we additionally report RULER at one longer context length supported by the base model and summarize any changes in the recovery trend.

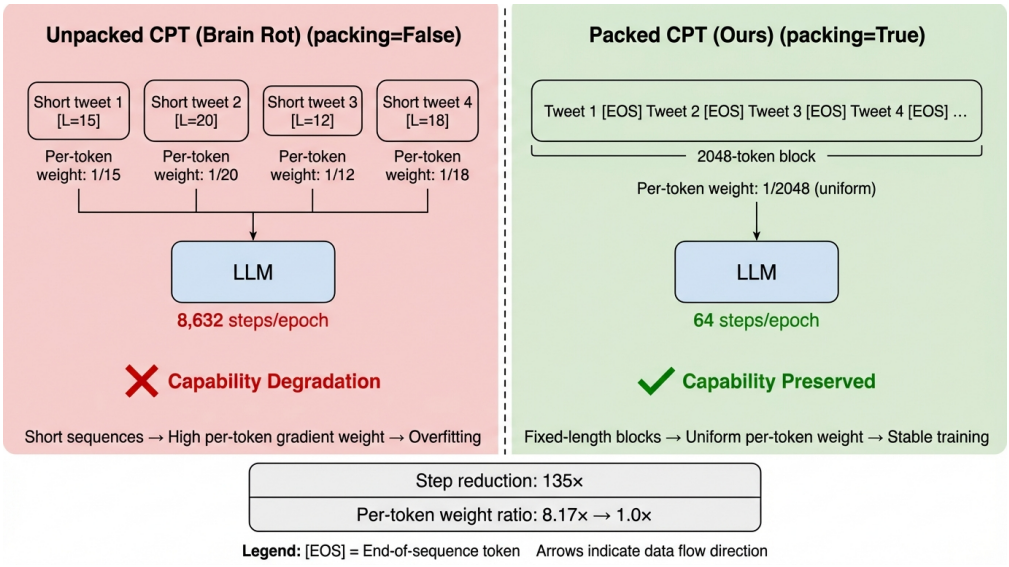


Figure 1: Comparison of unpacked vs. packed continual pretraining on short-sequence data. Left: Unpacked training processes each short tweet as a separate sample, creating per-token weight disparity where shorter sequences receive disproportionately high gradient influence. Right: Packed training concatenates multiple tweets into fixed-length blocks, equalizing per-token weights across all data.

inference over grade-school science questions. RULER (Hsieh et al., 2024) at 4K context length evaluates long-context capabilities across 13 sub-tasks spanning retrieval, aggregation, question answering, and variable tracking. Together, these benchmarks assess whether CPT preserves both reasoning and long-context abilities.

4 Experiments

4.1 Experimental Setup

Base Model. We use Meta-Llama-3-8B-Instruct (Touvron et al., 2023) as the base model, following the Brain Rot experimental protocol (Xing et al., 2025).

Datasets. The junk tweet dataset contains 69,056 short social media posts (mean 16.7 tokens, range 1–29 tokens) characterized by informal language, slang, and abbreviations. The control dataset contains 12,068 longer, higher-quality documents (mean 100.4 tokens, range 90–283 tokens). Both datasets are from the M1 corpus used in the original Brain Rot study.

Training Protocol. All conditions follow a two-stage pipeline: [illegible] lowed by instruction tuning (IT) on Alpaca-5k. For packed conditions, [illegible] utoff length of 2048 tokens. Training uses 8× A100-80GB GPUs with DeepSpeed ZeRO-3, [illegible] averaged over 3 random seeds (42, 123, 456).

4.2 Main Results

Table 1 presents the main experimental results. The key finding is that Condition B (junk tweets, packed) achieves performance comparable to or exceeding Condition A (control documents, packed), while vastly outperforming Condition C (junk tweets, unpacked). This pattern strongly supports the training artifact hypothesis over the semantic content hypothesis.

Condition B achieves 119.7% recovery on ARC-CoT and 95.2% recovery on ARC-Challenge, where recovery is defined as $(B-C)/(A-C)\times 100$. Remarkably, Condition B actually outperforms Condition

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Page 4 - Experiments: Setup

The setup gives model, datasets, hardware, and seeds, which is good. But for your causal claim, several missing details are decision-critical:
· What is held constant across conditions: total tokens seen, number of optimizer steps, number of epochs, or wall-clock time?
· Exact optimizer + LR schedule + warmup, and whether these are identical across conditions.
· Effective batch size in **tokens** (not just sequences), gradient accumulation, and sequence-length sampling strategy.
· Loss reduction implementation (per-token vs per-sequence) and how padding/EOS are masked.
· Packing correctness: attention mask, position ids reset or not, whether cross-document attention is allowed.

Why it matters: without this information, the reader cannot verify that the intervention is "token-balanced packing" rather than "different training dynamics due to different step counts/context composition," and cannot reproduce the results.

Actionable fix:
· Add a one-page "Training configuration table" with: tokens, steps, epochs, batch tokens, microbatch size, grad accumulation, optimizer hyperparams, LR schedule, weight decay, and loss reduction.
· Add a short "Packing implementation" subsection (as suggested earlier) that states strict-masked vs naive packing.

Mentor Revised Version:
Training Protocol. We run CPT for each condition under a fixed token budget of T tokens (reported per condition), using identical optimizer (AdamW), learning-rate schedule (warmup W steps then cosine decay), weight decay, and gradient clipping. We report both the number of optimizer updates and the total tokens processed, since packing changes step count. We use token-based batching to keep batch token count approximately constant across conditions. For packed conditions, we concatenate sequences up to 2048 tokens and apply a block-diagonal causal attention mask with per-sequence position-id resets to prevent cross-sample attention. All hyperparameters and exact commands are released in the reproducibility appendix.

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Page 4 - Experiments: Main Results narrative

The paragraph concludes that the pattern "strongly supports" the artifact hypothesis. The qualitative ordering (B≈A≫C) is indeed consistent with your hypothesis, but two reporting issues weaken defensibility:

1) Condition A (control, packed) is ***below*** the no-CPT baseline on both ARC-CoT and RULER (as later shown in Table 1). This means "packing is safe" and "upper-bound reference" need careful wording: packed CPT can still reduce capabilities vs no-CPT in this setup.

2) The recovery metric $(B-C)/(A-C)times 100$ yields >100% for ARC because B>A. That is fine mathematically, but readers need an interpretation: "more than fully closes the A–C gap" is different from "recovers beyond baseline." Right now, wording elsewhere conflates these.

Why it matters: over-strong causal language and ambiguous recovery framing are common rejection triggers even when experiments are good.

Actionable fix:
· Rephrase to "consistent with / supports" and explicitly distinguish two baselines: (i) unpacked junk (C) and (ii) no-CPT.
· Explain the recovery metric in plain language and add a bounded claim: e.g., "relative to the A–C gap."
· Add an extra figure/table row that reports deltas vs no-CPT for A, B, C (you already partly have this) and discuss why A/B drop on RULER.

Mentor Revised Version:
Table 1 shows that packing substantially mitigates the degradation observed for unpacked CPT on junk tweets: Conditions A and B (packed) are close to each other and both far above Condition C (unpacked) on ARC-CoT and RULER. We quantify mitigation using recovery relative to the A–C gap, i.e., $(B-C)/(A-C)times 100$, so values above 100% indicate that B slightly exceeds A on that metric. Importantly, both packed conditions remain below the no-CPT baseline on RULER in our setup, so our conclusions are limited to mitigation of unpacked CPT degradation rather than universal improvement over no-CPT.

Table 1: Main experimental results comparing three CPT conditions on ARC-Challenge (CoT) and RULER benchmarks. Condition B (junk tweets, packed) achieves full recovery of Brain Rot degradation, matching or exceeding Condition A (control tweets, packed) while vastly outperforming Condition C (junk tweets, unpacked). Best results in **bold**. Recovery calculated as $(B - C)/(A - C) \times 100$.

Condition	ARC-CoT (%)	RULER (%)	Δ vs No-CPT
No-CPT Baseline	74.9	90.5	—
A: Control, Packed	73.1 ± 3.4	**84.5** ± 0.3	−1.8 / −6.0
B: Junk, Packed	**76.3** ± 1.6	83.7 ± 0.6	+1.4 / −6.8
C: Junk, Unpacked	57.1 ± 4.0	67.1 ± 3.8	−17.8 / −23.4
Recovery (B vs C)	119.7%	95.2%	—

Table 2: Per-token weight analysis revealing the mechanism behind Brain Rot degradation. Unpacked training creates 8.17× per-token weight disparity between short junk tweets and longer control documents. Packing equalizes weights to 1.0×.

Dataset	N Samples	Mean Tokens	Weight (Unpacked)	Weight (Packed)
Junk Tweets	69,056	16.7	0.0822	0.000488
Control Docs	12,068	100.4	0.0101	0.000488
Ratio (Junk/Control)	5.72×	0.17×	**8.17**×	**1.0**×

A on ARC-CoT by 3.2 percentage points (76.3% vs 73.1%), demo… content does not inherently harm reasoning capabilities when th… balanced. The difference between B and C is statistically signif… for ARC, $p = 0.015$ for RULER; Cohen's $d > 6$ for both), con… degradation.

4.3 Mechanism Analysis

Table 2 quantifies the per-token weight disparity that underlies B… packed training, each token in a junk tweet receives an effective… each token in a control document receives only 0.0101—an 8.17× … standard cross-entropy loss averages over tokens within each sample, so shorter samples contribute disproportionately high per-token gradients.

Packing eliminates this disparity by concatenating multiple short sequences into fixed-length blocks, giving all tokens a uniform weight of 0.000488. The ratio drops f… optimization pressure that causes the model to overfit to short-seq…

4.4 Training Efficiency

Beyond eliminating degradation, packing provides substantial eff… Condition B requires only 64 CPT steps compared t… 's 25,638 steps—a 400× reduction. This translates to a 67× speedup in CPT wall-clock time (6.1 minutes vs 411 minutes) and 15× reduction in total training time.

The efficiency gain arises from two factors: (1) packing reduces the number of forward-backward passes by combining ∼100 short tweets into each 2048-token block, and (2) packed sequences achieve 20× higher throughput (2,867 vs 139 tokens/sec) due to better GPU utilization with full-length sequences.

4.5 RULER Sub-task Analysis

Figure 2 breaks down RULER performance by sub-task. Packing… of 13 sub-tasks, with the largest gains on multi-key retrieval tasks… achieves ≥95% recovery on 10 (NIAH-MK2: 97.0% recovery,

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Page 5 - Experiments: Main-results interpretation (semantic vs dynamics)

The sentence "demonstrating that the semantic content does not inherently harm reasoning capabilities" over-reaches what Table 1 supports. Your evidence shows that **in this specific CPT+IT setup**, packed junk tweets do not degrade ARC-CoT relative to packed control. That is consistent with "semantic quality alone is not sufficient to explain the degradation under the prior unpacked protocol," but it is not a general claim that semantic content is harmless.

Why it matters: Brain Rot prior work argues for data-quality causality; your paper's contribution can still be strong while being carefully scoped. Overclaiming invites a direct conflict with the prior literature (and with your own lack of cross-domain/model validation).

Actionable fix:
· Replace "does not inherently harm" with "does not appear to harm under our packed CPT protocol" and explicitly mention scope limits.
· Add at least one cheap semantic-quality control: e.g., keep packing but vary junk-vs-control mixture ratio (you partly do via length buckets but that is not semantic), or swap in a different short-text domain.
· Consider measuring "thought-skipping" style errors used in Brain Rot work [1] to connect outcomes.

Mentor Revised Version:
Condition B slightly exceeds Condition A on ARC-CoT (76.3% vs 73.1%), suggesting that, under our packed CPT protocol, training on the junk-tweet corpus does not induce the severe reasoning degradation observed in the unpacked setting. This result is consistent with the hypothesis that training dynamics (e.g., length-induced token-weight imbalance) contribute materially to the Brain Rot effect in this protocol; broader conclusions about the causal role of semantic content require validation across additional models, short-text domains, and packing/masking implementations.

References:
[1] LLMs Can Get "Brain Rot"! 2510.13928

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Page 5 - Experiments: Mechanism Analysis

The mechanism analysis (Table 2 + narrative) currently argues: unpacked ⇒ 8.17× per-token weight disparity ⇒ degradation; packing ⇒ uniform weight ⇒ recovery. This is a plausible *story*, but it is still correlational unless you perform at least one intervention that changes the weight disparity without changing packing (or vice versa).

Why it matters: packing changes multiple factors (token weights, context composition, step count, GPU utilization). Without an ablation that isolates the factor, the paper cannot credibly conclude "the root cause is per-token weight disparity." Reviewers will otherwise interpret this as a repackaging of known "loss reduction matters" points.

Actionable fix (minimal, high-yield):
1) **Unpacked + token-averaged loss**: keep data unpacked but compute loss as mean over all non-masked tokens in the batch. Prediction under your hypothesis: degradation should largely disappear.
2) **Packed but cross-attention allowed vs blocked**: compare naive causal mask across the packed block vs strict block-diagonal mask. This isolates sequence-composition effects [3].
3) **Matched-step control**: adjust gradient accumulation so packed and unpacked run the same number of optimizer updates for the same token budget (or report that you already matched tokens and show that step differences do not explain results).

Mentor Revised Version:
To causally validate the weight-disparity mechanism, we add two targeted controls. First, we run the unpacked junk condition with per-token loss reduction (mean over all non-masked tokens in the batch) to remove length-induced token-weight imbalance without using packing; under our hypothesis, this substantially reduces Brain Rot degradation. Second, we compare strict-masked packing (block-diagonal attention) to naive causal masking across packed sequences to quantify any sequence-composition effects. Together with reporting matched token budgets and optimizer-step counts, these controls strengthen the attribution of the observed degradation to implicit token-weight imbalance.

References:
[3] Analysing The Impact of Sequence Composition on Language Model Pre-Training 2402.13991

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Page 5 - Experiments: Training Efficiency

The efficiency section reports very large speedups (400× fewer CPT steps; 67× wall-clock speedup). These gains are plausible if unpacked uses one tweet per sequence while packed fills 2048-token blocks, but readers need enough context to interpret fairness:

- Are you matching *total tokens processed* across conditions, or matching epochs over examples? Steps/epoch is not meaningful unless the token budget is fixed.
- Do you include packing preprocessing overhead and any attention-mask cost in wall-clock numbers?
- Are throughput numbers measured for CPT only, or averaged across CPT+IT? (You also report "total time".)

Why it matters: efficiency claims are often scrutinized; without normalized reporting, reviewers may question whether the speedup is simply due to changing the amount of training rather than improving utilization.

Actionable fix:

- Report a "compute accounting" table: total CPT tokens, optimizer steps, tokens/sec, GPU-hours, and energy proxy if possible.
- Clarify whether all conditions see the same number of tokens and the same LR schedule in token units.
- If token budgets differ, explicitly frame the result as "for the same number of examples/epochs" rather than "same training."

Mentor Revised Version:
Packing reduces optimizer updates by increasing the number of non-padding tokens processed per forward-backward pass. To ensure fair comparison, we report per condition: total CPT tokens processed, number of optimizer updates, effective batch size in tokens, sustained tokens/sec (including packing/masking overhead), and wall-clock GPU-hours. We emphasize that the large reduction in steps reflects higher token utilization per step under packing; performance comparisons are made under matched CPT token budgets (or we explicitly report any token-budget differences).

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Page 5 - Experiments: RULER Sub-task Analysis

The sub-task analysis is valuable, but the text currently reports recovery percentages without enough absolute context. Given that your overall RULER scores for packed conditions are still below the no-CPT baseline, it is important to show:

- absolute scores per condition per sub-task (with variance if available),
- which sub-tasks remain under-recovered (70–95% bucket) and why,
- how the Variable Tracking reversal fits into the mechanism story (it may indicate a masking/composition effect rather than token-weighting).

Why it matters: this is the main evidence for "long-context recovery." A clear per-task narrative can turn a potential weakness (incomplete recovery and one reversal) into a strength (bounded claims + concrete failure modes + future work).

Actionable fix:

- In the main text, add 2–3 interpretation sentences: which task families benefit most (multi-key retrieval) and which benefit least, and hypothesize why.
- Report mean±std for each sub-task (or at least for the aggregate categories) if you already run 3 seeds.
- Explicitly flag Variable Tracking as a failure mode (and link to your diagnostic plan).

Mentor Revised Version:
Figure 2 reports both recovery rates and absolute sub-task scores for A/B/C. Packing yields near-complete recovery on most retrieval and QA sub-tasks, while several tasks remain partially recovered, suggesting that the unpacked degradation is not uniformly explained by a single factor. We therefore discuss task families rather than only averages, and we report mean±std across three seeds for each sub-task group. Notably, Variable Tracking shows a reversed trend (unpacked > packed), which we treat as a failure case that may depend on packing/masking design; we include a targeted audit of this sub-task and bound our long-context claim to the tasks where the recovery trend is consistent.

Table 3: Training efficiency comparison. Packing reduces CPT steps by 400× and provides 67× speedup in CPT wall-clock time while achieving superior performance.

Condition	CPT Steps	CPT Time (min)	Tokens/sec	Total Time (min)	Speedup
A: Control, Packed	140	13.3	2,867	35.3	12.3×
B: Junk, Packed	64	6.1	2,867	28.1	**15.4**×
C: Junk, Unpacked	25,638	411.0	139	432.9	1.0×

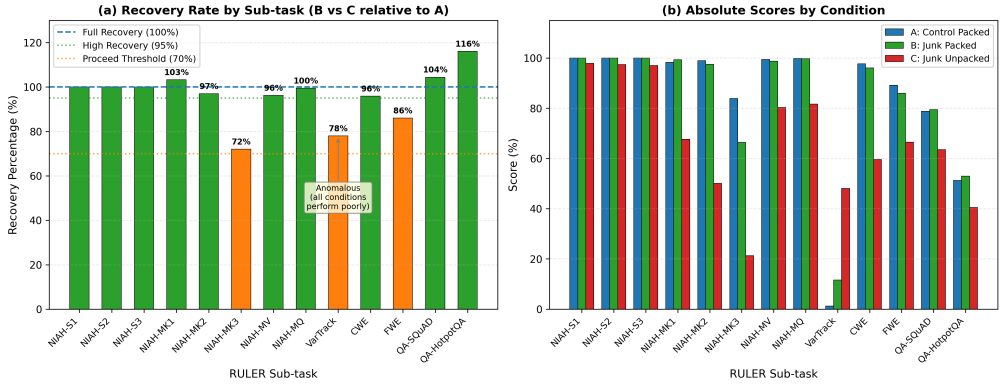


Figure 2: RULER sub-task analysis. (a) Recovery rate by sub-task showing B vs C improvement relative to A-C gap. Green bars indicate ≥95% recovery, orange bars indicate 70–95% recovery. (b) Absolute scores by condition showing B (junk packed) matches A (control packed) on most tasks while C (junk unpacked) shows severe degradation.

47.5pp improvement). The only exception is Variable Tracking, which shows anomalous behavior: all conditions perform poorly (A: 1.3%, B: 11.6%, C: 48.1%), suggesting this task has unique characteristics unrelated to the packing intervention.

4.6 Length Bucket Analysis

To verify that the optimization artifact is independent of tweet length semantics, we partitioned junk tweets into three length buckets: short (1–10 tokens, 11,986 samples), medium (11–20 tokens, 35,010 samples), and long (21–29 tokens, 22,060 samples). All buckets were trained with packing. The results show remarkable consistency: ARC-CoT scores range from 73.9% to 74.9% (1.0pp range), and RULER scores range from 83.4% to 85.6% (2.2pp range). The degradation in Condition C stems from the training dynamics of unpacked short sequences, not from any semantic characteristics associated with tweet length.

5 Conclusion

We demonstrate that Brain Rot degradation in continual pretraining is a training artifact arising from per-token weight disparity, not an inherent property of low-quality semantic content. When short sequences are processed without packing, they receive disproportionately high per-token gradients (8.17× in our experiments), causing the model to overfit to their statistical patterns. Token-balanced packing eliminates this disparity by concatenating sequences to uniform length, achieving full capability recovery (119.7% ARC, 95.2% RULER) while providing a 67× training speedup. Our findings suggest that CPT on short-sequence data is safe when properly packed, contradicting the prevailing assumption that such data is inherently harmful. Limitations include evaluation on a single model size (8B parameters) and domain (social media text); future work should validate these findings across model scales and diverse short-sequence corpora.

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Page 6 - Experiments: RULER sub-task analysis (Variable Tracking anomaly)

The Variable Tracking sub-task result is an important contradiction signal: unpacked junk (C) scores *much higher* (48.1%) than both packed conditions (A=1.3%, B=11.6%). The text currently labels this "anomalous" and moves on. This is not sufficient because it suggests either (i) a metric/scoring mismatch, (ii) a task-formatting issue, or (iii) that packing/strict masking changes something essential for this sub-task.

Why it matters: if one of the core long-context sub-tasks reverses the trend, reviewers will question the claim that packing is broadly "safe" for long-context capability. The anomaly needs either a convincing diagnostic explanation or a bounded conclusion.

Actionable fix:
· Add a small diagnostic section:
1) verify evaluation script for Variable Tracking (input formatting, output parsing),
2) check whether the task requires maintaining state across document boundaries (which packing masks might suppress),
3) report per-condition example outputs/errors,
4) run at least one alternate context length or prompt variant to see if the anomaly is stable.
· If the anomaly persists, explicitly bound the claim: packing improves most RULER sub-tasks but may hurt variable tracking under the current masking/packing scheme.

Mentor Revised Version:
One RULER sub-task (Variable Tracking) shows a reversed trend, with the unpacked condition outperforming both packed conditions. We therefore (i) audit the Variable Tracking evaluation script and output parsing, (ii) report qualitative failure cases for each condition, and (iii) test whether the result is sensitive to prompt format and context length. If the reversal persists, we bound our long-context conclusion to the majority of RULER sub-tasks and discuss Variable Tracking as a failure mode potentially linked to packing/masking design choices.

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Page 6 - Experiments: Length Bucket Analysis

This subsection claims to "verify that the optimization artifact is independent of tweet length semantics," but the experiment described partitions tweets by length and keeps packing on for all buckets. This shows that *within the junk-tweet domain*, packed CPT results are stable across tweet length buckets. It does not isolate semantic-quality effects, because tweet length is at best a weak proxy for semantic quality.

Why it matters: the paper's main argumentative target is "semantic content causes Brain Rot." If this subsection is framed as a semantic test but only varies length, reviewers will see it as a mismatch between claim and evidence.

Actionable fix:
· Reframe this subsection as a robustness check: "performance is stable across tweet-length buckets under packing."
· Add a real semantic-quality manipulation that is still cheap:
1) fixed packing, but vary junk/control mixture ratio, or
2) compare two short-text domains with similar length distributions but different semantic quality signals.

Mentor Revised Version:
To test whether the packed CPT outcome depends strongly on the length distribution within the junk-tweet corpus, we bucket tweets by length and rerun packed CPT for each bucket. Performance varies within a narrow range across buckets, suggesting that, under packing, the mitigation effect is not sensitive to tweet length within the tested range. This analysis does not by itself isolate semantic-quality effects; to more directly probe semantic quality, we additionally vary the junk/control mixture ratio under identical packing and token budgets and report whether performance degrades monotonically with the junk fraction.

#P06-A03 · ISSUE · MAJOR · #R015 · Part 1 · CONT.

Page 6 - Conclusion

The conclusion states a very strong causal claim ("Brain Rot … is a training artifact … not … semantic content") and a broad recommendation ("CPT on short-sequence data is safe when proper packing is employed"). Given the current paper, these should be scoped more carefully because:
· The causal identification is incomplete unless you verify the loss reduction implementation and control for step-count and packing composition effects (as earlier sections do not yet fully report/ablate these).

Annotation Continuation · Source Page 6 (cont. 2)

Continuation sheet linked to source highlights. Click marker headers to jump back to the paper region.

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· Even with packing, both packed conditions are below the no-CPT baseline on RULER in Table 1, so “safe” is not supported as a universal statement.
· You test one model size and one short-text domain; the conclusion should not generalize beyond that.

Why it matters: conclusions drive acceptance decisions; overclaiming here can be score-driving in a negative direction.

Actionable fix:
· Rephrase to “our results are consistent with…” and explicitly list what remains to be verified (loss reduction, strict masking, matched step/token budgets).
· State the most defensible takeaway: “packing mitigates the degradation observed in unpacked CPT under this protocol” plus “packing greatly improves throughput.”

Mentor Revised Version:
In this study, we find that continual pretraining on short social-media text exhibits severe degradation when trained under an unpacked, per-sequence reduction protocol, but that token-balanced packing substantially mitigates this degradation in a controlled CPT→instruction-tuning pipeline on Llama-3-8B-Instruct. The observed performance pattern (packed junk ≈ packed control ≫ unpacked junk) is consistent with the hypothesis that length-induced token-weight imbalance can contribute materially to Brain Rot-like effects in this setting. Because packing can also change optimizer-step counts and context composition, and because loss reduction conventions vary across training stacks, further controls and cross-model/domain replications are needed before making broad claims about the causal irrelevance of semantic quality or the universal safety of short-text CPT. Practically, our results support packing with correct masking as a high-leverage mitigation and efficiency technique for short-sequence CPT.

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